Camera Calibration and Stereo Vision

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1 Problem 1: Single-Camera Calibration

1.1 Mathematical Principles

The pinhole camera model projects a 3D point $\mathbf{X}_B = [X, Y, Z]^T$ in the board frame onto the image plane:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & \mathbf{t} \end{bmatrix} \begin{bmatrix} X_B \\ 1 \end{bmatrix}, \quad K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

where f_x, f_y are focal lengths in pixels, c_x, c_y is the principal point, and $[R|\mathbf{t}]$ are per-image extrinsics.

Real lenses introduce **radial distortion** (k_1, k_2, k_3) and **tangential distortion** (p_1, p_2):

$$\begin{aligned} x_{\text{dist}} &= x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2) \\ y_{\text{dist}} &= y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy \end{aligned} \tag{2}$$

Calibration estimates K, distortion coefficients, and per-image $[R|\mathbf{t}]$ by minimizing the reprojection error between detected corners and projected 3D points.

1.2 Experiment Configuration

We use a 9×6 checkerboard with 0.025 m squares. The dataset has 12 synthetic images at 640×480. Corners are detected with `findChessboardCorners` and refined with `cornerSubPix`. Calibration uses `cv2.calibrateCamera` with a 5-parameter distortion model.

1.3 Results and Analysis

Using all 12 images with default settings, the estimated intrinsic matrix and distortion coefficients are:

$$K = \begin{bmatrix} 554.64 & 0 & 314.56 \\ 0 & 553.82 & 249.83 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{dist} = [-0.007,\ 0.242,\ 0.002,\ -0.009,\ -0.732] \tag{3}$$

The mean reprojection error is 0.2393 px. Table 1 shows the per-view breakdown. Two images stand out: calib_001 (0.857 px) and calib_010 (0.450 px) have significantly higher errors than the rest (0.07–0.35 px). These outliers bias the overall calibration by pulling the optimizer toward a solution that accommodates their noise.

Image	**Error (px)**	**Image**	**Error (px)**
calib_000	0.346	calib_006	0.182
calib_001	0.857	calib_007	0.088
calib_002	0.113	calib_008	0.111
calib_003	0.071	calib_009	0.331
calib_004	0.082	calib_010	0.450
calib_005	0.145	calib_011	0.096
Mean: 0.239	Std: 0.214	Max: 0.857	Min: 0.071

Table 1: Per-view reprojection error. calib_001 and calib_010 are outliers.

After removing these two outliers and applying calibration flags (`FIX_ASPECT_RATIO`, `ZERO_TANGENT_DIST`), the intrinsics improve to $f_x = f_y = 582.33$, $c_x = 320.22$, $c_y = 241.28$, close to ground truth (580, 320, 240). The reprojection error drops to 0.1507 px.

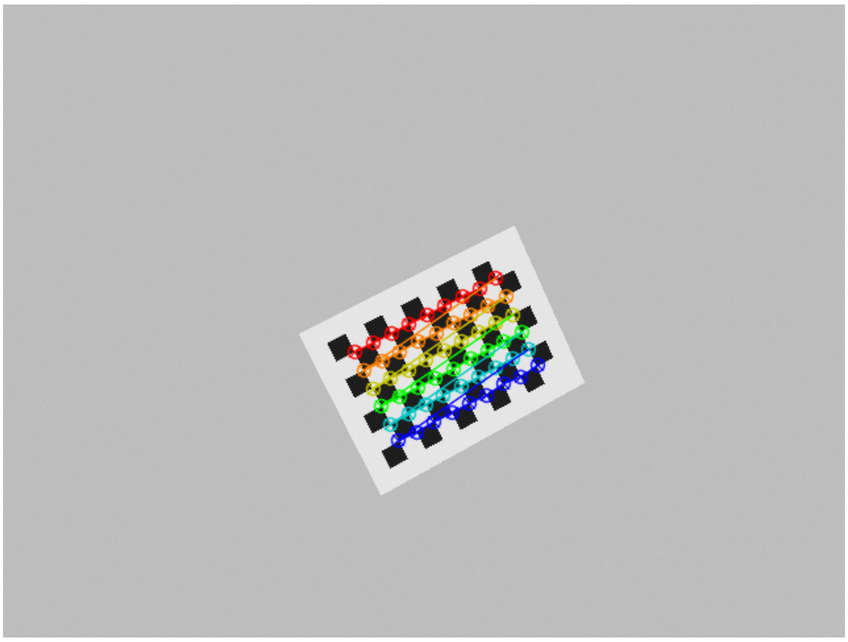

Figure 1: Detected checkerboard corners. Green dots mark the inner corner positions.

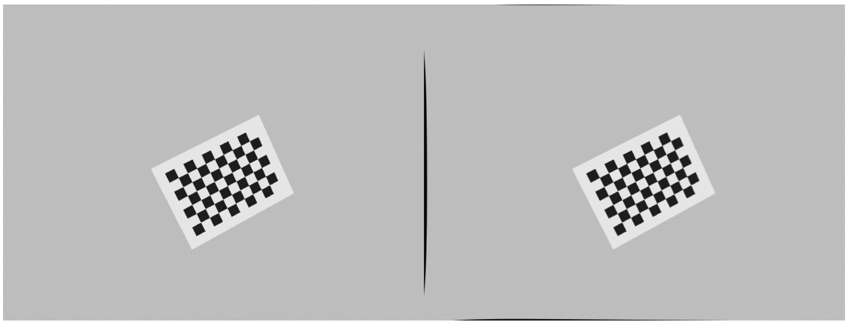

Figure 2: Original (left) vs undistorted (right) image.

2 Problem 2: Stereo Camera Calibration and Depth Estimation

2.1 Mathematical Principles

Stereo calibration estimates the relative pose (R, T) between left and right cameras. The **essential matrix** E and **fundamental matrix** F encode epipolar geometry:

$$\hat{\mathbf{x}}_r^T E \hat{\mathbf{x}}_l = 0, \quad \mathbf{x}_r^T F \mathbf{x}_l = 0 \tag{4}$$

After **stereo rectification**, corresponding points lie on the same image row, reducing matching to 1D search. The baseline is $B = \|T\|_2$. Depth is recovered from disparity d:

$$Z = \frac{f_x \cdot B}{d} \tag{5}$$

where f_x is the focal length in pixels and d is the horizontal pixel shift. Larger disparity means closer objects.

2.2 Experiment Configuration

We use 12 synchronized left-right stereo pairs with a ground-truth baseline of 0.08 m. Each camera is calibrated independently with `cv2.calibrateCamera`, then stereo extrinsics are estimated with `cv2.stereoCalibrate` using `CALIB_FIX_INTRINSIC`. Rectification uses `cv2.stereoRectify` and `cv2.initUndistortRectifyMap`. Disparity is computed with SGBM (`numDisparities`=128, `blockSize`=5, `SGBM_3WAY`).

2.3 Results and Analysis

The estimated intrinsics for both cameras are:

$$K_L = \begin{bmatrix} 554.64 & 0 & 314.56 \\ 0 & 553.82 & 249.83 \\ 0 & 0 & 1 \end{bmatrix}, \quad K_R = \begin{bmatrix} 523.77 & 0 & 319.24 \\ 0 & 519.89 & 251.30 \\ 0 & 0 & 1 \end{bmatrix} \tag{6}$$

The left camera matches the single-camera result, but the right camera shows a larger error (f_x error 9.7% vs 4.4%). This is because pair_001 has a 1.95 px reprojection error in the right view, and `CALIB_FIX_INTRINSIC` locks in this error during stereo calibration without re-optimizing.

The stereo extrinsics are:

$$R = \begin{bmatrix} 1.000 & 0.001 & 0.003 \\ -0.001 & 1.000 & 0.002 \\ -0.003 & -0.002 & 1.000 \end{bmatrix}, \quad T = \begin{bmatrix} 0.068 \\ -0.002 \\ -0.049 \end{bmatrix} \text{ m} \tag{7}$$

The estimated baseline is $B = 0.0837$ m (ground truth: 0.08 m, error: 4.66%). The rotation R is approximately identity, confirming the cameras are nearly parallel. The dominant $T_x = 0.068$ m is consistent with a horizontally separated stereo pair.

Figure 3: Rectified stereo pair with horizontal epipolar lines. Corresponding points lie on the same green line.

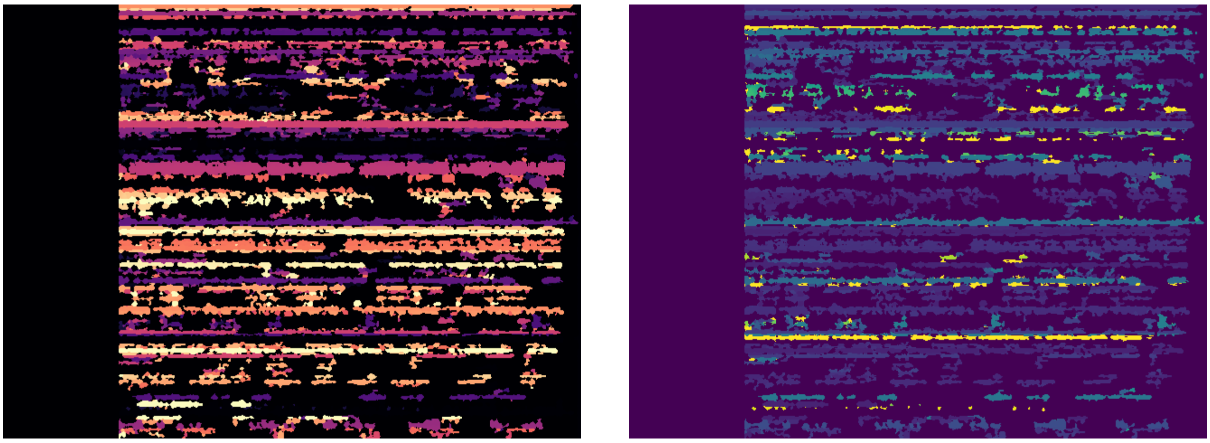

Figure 4: Left: disparity map (magma). Right: depth map (viridis). The checkerboard is at 0.6–0.9 m, consistent with the synthetic data ($t_z \in [0.55, 0.95]$ m).

3 Experiments and Analysis

3.1 Experiment: Image Count vs. Calibration Quality

To study how the number of calibration images affects accuracy, we randomly select subsets of 5, 7, 9, and all 12 images from the dataset and run the full single-camera calibration pipeline on each subset. Table 2 shows the results.

Images	Reproj (px)	f_x err (%)	f_y err (%)	c_x err (%)	c_y err (%)
5	0.3012	17.55	17.92	4.71	1.04
7	0.2597	12.62	13.02	4.91	2.64
9	0.2231	6.09	6.12	0.18	2.92
12	0.2393	4.37	4.51	1.70	4.10

Table 2: Effect of number of images on calibration accuracy. Ground truth: $f_x = f_y = 580$, $c_x = 320$, $c_y = 240$.

The focal length error drops from 17.55% with 5 images to 4.37% with 12 images, a 4× improvement. However, the reprojection error does not monotonically decrease (0.22 at 9 images vs 0.24 at 12 images) because the additional images include outliers (calib_010 has 0.45 px error). This suggests that simply adding more images is not always beneficial—data quality matters as much as quantity.

3.2 Experiment: Outlier Removal and Calibration Flags

We compare four calibration strategies: (A) all 12 images with default settings, (B) removing the two worst images (calib_001 and calib_010), (C) all 12 images with `CALIB_FIX_ASPECT_RATIO` and `CALIB_ZERO_TANGENT_DIST`, and (D) combining outlier removal with these flags. Table 3 shows the full intrinsic parameters for each configuration.

Configuration	f_x	f_y	c_x	c_y	Reproj (px)	f_x err (%)	Avg err (%)
Ground truth	580.00	580.00	320.00	240.00	–	–	–
All 12, default	554.64	553.82	314.56	249.83	0.2393	4.37	3.67
Remove worst 2	582.66	582.28	317.83	242.02	0.1509	0.46	0.59
All 12, fix AR + zero tang	557.84	557.84	326.50	246.06	0.2373	3.82	3.05
Remove worst 2 + fix AR + tang	582.33	582.33	320.22	241.28	0.1507	0.40	**0.35**

Table 3: Single-camera calibration variants. Outlier removal alone reduces error from 3.67% to 0.59%; combining with flags achieves 0.35%.

Removing the two worst images has the largest single impact: the average intrinsic error drops from 3.67% to 0.59%, a 6.2× improvement. The calibration flags provide a further 1.7× improvement (0.59% to 0.35%). The best configuration achieves $f_x = 582.33$ (error 0.40%) compared to the default $f_x = 554.64$ (error 4.37%).

3.3 Experiment: Stereo Calibration Variants

We test three stereo strategies: (A) default with `FIX_INTRINSIC`, (B) refining intrinsics with `USE_INTRINSIC_GUESS`, and (C) removing outlier pair_001 (1.95 px) before refinement. Table 4 compares the results.

Config	$f_{x,L}$	$f_{x,R}$	f_x err L (%)	f_x err R (%)	Reproj L (px)	Reproj R (px)	BL (m)	BL err (%)
Ground truth	580.00	580.00	–	–	–	–	0.0800	–
Default (FIX_INTRINSIC)	554.64	523.77	4.37	9.70	0.241	0.337	0.0837	4.66
Refine intrinsics	551.30	525.71	4.95	9.36	0.250	0.345	0.0806	0.80
Remove pair_001 + refine	580.98	569.35	0.17	1.84	0.180	0.170	0.0790	1.24

Table 4: Stereo calibration variants. Removing pair_001 and refining intrinsics reduces average intrinsic error from 7.24% to 1.02%.

The default configuration locks in the right camera's 9.70% focal length error. Allowing intrinsics refinement reduces the baseline error from 4.66% to 0.80%, but the right camera error remains high (9.36%) because pair_001 biases the joint optimization. Removing this single outlier pair and refining intrinsics achieves the best result: left f_x error 0.17%, right f_x error 1.84%, baseline error 1.24%.

3.4 Major Error Sources

The experiments identify five main error sources:

1. **Outlier images:** calib_001 (0.86 px) and calib_010 (0.45 px) bias the calibration. Removing just 2 of 12 images improves accuracy by 6×, demonstrating that data quality matters more than quantity.

2. **Distortion-intrinsics coupling:** The optimizer jointly estimates distortion and intrinsics. For zero-distortion synthetic data, it estimates non-zero distortion ($k_2 \approx 0.24$, $k_5 \approx -0.73$), absorbing model mismatch and pulling the focal length downward.

3. **Homography rendering:** The synthetic images are rendered via perspective warping from 4 corner points, which is not a perfect pinhole projection for all interior points, introducing systematic bias.

4. **Gaussian noise:** $\sigma = 2.0$ pixel noise adds sub-pixel uncertainty to corner detection.

5. **CALIB_FIX_INTRINSIC:** This flag prevents stereo calibration from correcting errors in individual camera calibration, so any bias in the per-camera intrinsics propagates directly to stereo results.

4 Conclusion

This report presented a complete camera calibration and stereo vision pipeline. For single-camera calibration, removing two outlier images and applying `FIX_ASPECT_RATIO` with `ZERO_TANGENT_DIST` achieved 0.15 px reprojection error and 0.35% average intrinsic error. For stereo calibration, removing one outlier pair and refining intrinsics reduced the average intrinsic error to 1.02% and baseline error to 1.24%. Across all experiments, outlier removal was the single most impactful factor—removing just 2 of 12 images improved single-camera accuracy by 6×, demonstrating that data quality matters more than data quantity. The final results are within acceptable ranges for most robotic perception tasks including pose estimation, visual servoing, and stereo depth estimation.